E-Commerce Store Analytics

Objective

This assignment shows how to clean, analyze, and build an interactive Excel dashboard from start to finish.

Dataset Description

The analysis is based on an E-commerce store dataset consisting of four interconnected tables: Customer_Dim (containing demographic details, geographic location, and loyalty tiers for 500 customers), Product_Dim (cataloguing 100 products across various categories, sub-categories, brands, manufacturing costs, and stock levels), Store_Dim (outlining 20 retail locations categorized by geographic region, city, and type like flagship, outlet, or online), and Sales_Fact (capturing 2,000 transaction records that link customers, products, and stores, complete with order dates, purchase quantities, unit prices, discounts, payment methods, and total sales amounts).

Column Description

Column Name	Description
Sales_ID	Unique transaction ID
Order_Date	The timestamp when the order was placed
Customer_ID	Unique customer ID
Product_ID	Unique product ID
Store_ID	Unique store ID
Quantity	Number of units purchased
Unit_Price	Selling price per single unit
Discount	Discount percentage applied to the sale
Payment_Type	Method of payment used
Total_Amount	Final calculated revenue (accounting for quantity, unit price, and discount)

Data Cleaning and Transformation

Customer Table Data Cleaning

Steps Using

- Convert into table using ctrl+t
- Find and replace
- PROPER(TRIM(C2))

- IF(OR(ISBLANK(H2),H2=0),"Unknown",H2)

Customer_ID	Name	Ag	Gende	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	Unknown
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	Unknown
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA	Gold

Sales Table Data Cleaning

Steps Using

- Convert to table
- IF(OR(ISBLANK(G2),G2=0),1,G2)→Quantity column
- IF(OR(ISBLANK(I2),I2=0),0,I2)→Discount column
- IF(ISBLANK(H2),K2/F2,H2)→Unit price
- IF(ISBLANK(J2),F2*G2*(1-H2),J2)→Total Amount

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	₹ 218.04	0.09	Credit Card	₹ 198.42
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	₹ 104.47	0.27	PayPal	₹ 228.79
SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	₹ 914.05	0.25	Credit Card	₹ 2,742.15
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	₹ 921.51	0.16	COD	₹ 774.07
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	₹ 745.30	0.18	Credit Card	₹ 1,833.44
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	₹ 147.01	0.19	COD	₹ 357.23
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	₹ 785.31	0.15	PayPal	₹ 1,335.03
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	₹ 342.39	0.23	PayPal	₹ 1,054.56
SALE8	04-05-2025	CUST1187	PROD99	STORE5	1	₹ 448.25	0.07	PayPal	₹ 1,250.62
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	₹ 873.17	0.23	PayPal	₹ 1,344.68
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	₹ 278.63	0.07	Credit Card	₹ 259.13
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	₹ 286.92	0.17	Credit Card	₹ 952.57
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	₹ 644.58	0.01	Credit Card	₹ 2,552.54
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	₹ 316.90	0.3	PayPal	₹ 443.66
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	₹ 403.47	0.29	PayPal	₹ 1,145.85
SALE15	10-12-2024	CUST1435	PROD36	STORE9	1	₹ 871.34	0.04	COD	₹ 2,509.46
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	₹ 451.98	0.25	Credit Card	₹ 677.97
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	₹ 502.69	0.23	Credit Card	₹ 774.14
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	₹ 559.23	0.25	COD	₹ 838.85
SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	₹ 445.61	0.04	COD	₹ 1,782.45
SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	₹ 986.04	0.27	COD	₹ 2,159.43
SALE21	04-06-2024	CUST1396	PROD43	STORE18	1	₹ 589.56	0.22	Credit Card	₹ 459.86
SALE22	12-04-2024	CUST1476	PROD69	STORE10	3	₹ 979.42	0.03	PayPal	₹ 2,850.11
SALE23	23-06-2024	CUST1368	PROD12	STORE8	1	₹ 719.28	0.08	Credit Card	₹ 661.74

Product Table Data Cleaning

Steps Using

- Convert to table
- Using find and replace to maintain consistency in product id
- =IF(ISBLANK(G2),AVERAGE(G2:G101),G2)→Cost and Stock column

Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	₹112.42	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	₹342.56	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc	₹254.62	352
PROD3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	₹364.67	452
PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	₹306.46	487
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	₹261.00	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	₹312.80	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	₹218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	₹439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	₹495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	₹298.86	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	₹449.40	271
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	₹115.82	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	₹242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowman	₹48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	₹465.34	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	₹337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	₹144.53	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	₹245.78	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	₹21.34	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	₹131.51	264
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	₹123.46	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodriguez	₹484.62	432
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos	₹248.82	360
PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	₹166.06	340
PROD25	Women's Scarf Set	Clothing	Accessories	Brown and Sons	₹174.02	268
PROD26	Wireless Bluetooth Speaker	Electronics	Audio	Reed-Schmidt	₹322.41	213
PROD27	Decorative Floor Vase	Home	Decor	Khan, Jones and Paul	₹250.02	378
PROD28	Sports Stopwatch Timer	Sports	Accessories	Burns-Hansen	₹38.76	137
PROD29	Fabric Sofa Throw Blanket	Home	Decor	Hoffman Group	₹218.48	276
PROD30	Stainless Steel Cutlery Set	Home	Kitchen	Stevens, Fernandez and Martin	₹19.52	57

Customer_Dim | Product_Dim | Store_Dim | Sales_Fact

Summary Statistics Using Toolpak

	Unit_Price		*Total_Amount*
Mean	513.5701533	Mean	1087.197349
Standard Error	6.313649626	Standard Error	18.3272372
Median	522.43	Median	868.7182
Mode	871.34	Mode	#N/A
Standard Deviation	282.354995	Standard Deviation	819.6189642
Sample Variance	79724.34321	Sample Variance	671775.2464
Kurtosis	-1.193225506	Kurtosis	0.045826153
Skewness	-0.034754787	Skewness	0.87904704
Range	979.8	Range	3889.7838
Minimum	20.09	Minimum	18.459
Maximum	999.89	Maximum	3908.2428
Sum	1027140.307	Sum	2174394.698
Count	2000	Count	2000

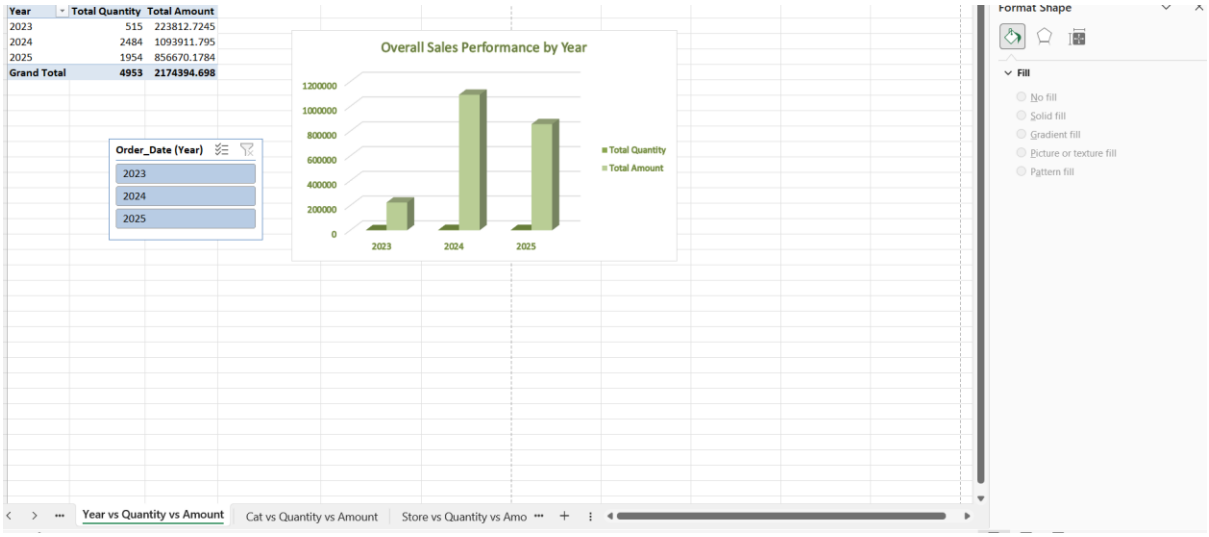
Use the XLOOKUP Functions to Combine the Tables

XLOOKUP([@[Product_ID]],Table3[Product_ID],Table3[Product_Name],0)

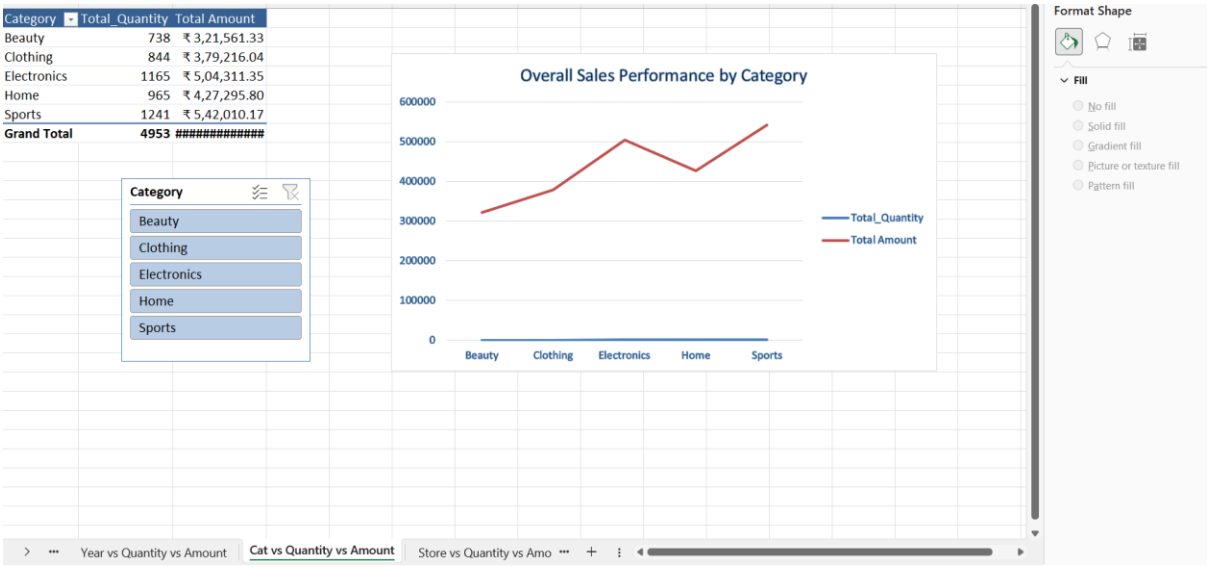
XLOOKUP([@[Customer_ID]],Table2[Customer_ID],Table2[Name],"Unknown",0)

Sales_ID	Order_Date	Customer_ID	Customer_Name	Product_ID	Product_Name	Store_ID	Quantity	Unit_Price	Discount	Payment_Ty	tal_Amou
SALE0	07-07-2024	CUST1409	Sarah Smith	PROD97	Cricket Bat	STORE3	1	₹ 218.04	0.09	Credit Card	₹ 198.42
SALE1	25-12-2023	CUST1467	Michael Foley	PROD74	Bluetooth Car Audio System	STORE17	3	₹ 104.47	0.27	PayPal	₹ 228.79
SALE2	03-06-2024	CUST1056	Christopher Sutton	PROD49	Men's Roll-On Perfume	STORE8	4	₹ 914.05	0.25	Credit Card	₹ 2,742.15
SALE3	29-11-2023	CUST1346	John Jordan	PROD76	Wireless Mouse	STORE8	1	₹ 921.51	0.16	COD	₹ 774.07
SALE4	06-01-2024	CUST1493	Amanda Powell	PROD59	Men's Hair Styling Gel	STORE5	3	₹ 745.30	0.18	Credit Card	₹ 1,833.44
SALE5	26-07-2024	CUST1116	Anthony Phillips	PROD15	4K Home Projector	STORE7	3	₹ 147.01	0.19	COD	₹ 357.23
SALE6	12-07-2025	CUST1016	John Scott	PROD56	Premium Blanket Throw	STORE4	2	₹ 785.31	0.15	PayPal	₹ 1,335.03
SALE7	18-11-2023	CUST1152	Patrick Roberts	PROD70	Kitchen Storage Rack	STORE8	4	₹ 342.39	0.23	PayPal	₹ 1,054.56
SALE8	04-05-2025	CUST1187	Dr. Suzanne Bullock	PROD99	Stainless Steel Cookware Set	STORE5	1	₹ 448.25	0.07	PayPal	₹ 1,250.62
SALE9	31-07-2025	CUST1464	David Pierce	PROD33	Modern Wall Clock	STORE0	2	₹ 873.17	0.23	PayPal	₹ 1,344.68
SALE10	13-02-2024	CUST1065	Michael Thomas	PROD24	Foldable Camping Chair	STORE14	1	₹ 278.63	0.07	Credit Card	₹ 259.13
SALE11	24-05-2025	CUST1320	Jessica Gonzalez	PROD39	Portable Power Bank	STORE1	4	₹ 286.92	0.17	Credit Card	₹ 952.57
SALE12	24-08-2024	CUST1237	Wesley Short	PROD88	Sports Wrist Sweatbands	STORE4	4	₹ 644.58	0.01	Credit Card	₹ 2,552.54
SALE13	06-08-2025	CUST1443	Adam Castro	PROD34	Gaming Mechanical Keyboard	STORE16	2	₹ 316.90	0.3	PayPal	₹ 443.66
SALE14	21-06-2025	CUST1348	Sherry Cabrera	PROD24	Foldable Camping Chair	STORE15	4	₹ 403.47	0.29	PayPal	₹ 1,145.85
SALE15	10-12-2024	CUST1435	Lindsay Beasley	PROD36	Women's Summer Dress	STORE9	1	₹ 871.34	0.04	COD	₹ 2,509.46
SALE16	13-09-2025	CUST1276	Arthur Warren	PROD65	Adjustable Dumbbells 20kg	STORE19	2	₹ 451.98	0.25	Credit Card	₹ 677.97
SALE17	11-12-2024	CUST1079	Molly Gardner	PROD27	Decorative Floor Vase	STORE4	2	₹ 502.69	0.23	Credit Card	₹ 774.14
SALE18	13-08-2024	CUST1094	David Patterson	PROD39	Portable Power Bank	STORE4	2	₹ 559.23	0.25	COD	₹ 838.85
SALE19	18-03-2024	CUST1239	Maria Hill	PROD60	Luxury Perfume Spray	STORE13	4	₹ 445.61	0.04	COD	₹ 1,782.45
SALE20	05-07-2025	CUST1144	Justin Cross	PROD72	Professional Jump Rope	STORE17	3	₹ 986.04	0.27	COD	₹ 2,159.43
SALE21	04-06-2024	CUST1396	Alexander Byrd	PROD43	USB External Hard Drive	STORE18	1	₹ 589.56	0.22	Credit Card	₹ 459.86
SALE22	12-04-2024	CUST1476	Elizabeth Foster	PROD69	Men's Sports Shorts	STORE10	3	₹ 979.42	0.03	PayPal	₹ 2,850.11
SALE23	23-06-2024	CUST1368	Miranda King	PROD12	Hair Strengthening Shampoo	STORE8	1	₹ 719.28	0.08	Credit Card	₹ 661.74
SALE24	08-09-2025	CUST1186	Jennifer Pruitt	PROD67	Smart LED Light Strip	STORE6	3	₹ 77.60	0.07	Credit Card	₹ 216.50
SALE25	17-09-2024	CUST1381	Erika Padilla	PROD5	Men's Leather Belt	STORE3	4	₹ 852.71	0.14	PayPal	₹ 2,933.32
SALE26	03-01-2024	CUST1213	Stuart Johnson	PROD98	Portable Bluetooth Speaker	STORE8	4	₹ 156.16	0	Credit Card	₹ 599.65
SALE27	22-02-2025	CUST1200	Anna Perez	PROD77	WiFi Mesh Router	STORE8	3	₹ 815.49	0.17	Credit Card	₹ 2,030.57
SALE28	23-09-2025	CUST1148	Theresa Campbell	PROD13	USB-C Fast Charger	STORE2	2	₹ 74.21	0.11	PayPal	₹ 132.09
SALE29	11-01-2024	CUST1348	Sherry Cabrera	PROD99	Stainless Steel Cookware Set	STORE3	1	₹ 451.67	0.15	Credit Card	₹ 1,535.68
SALE30	18-10-2023	CUST1028	Joshua Ryan	PROD5	Men's Leather Belt	STORE19	1	₹ 429.38	0.13	PayPal	₹ 373.56

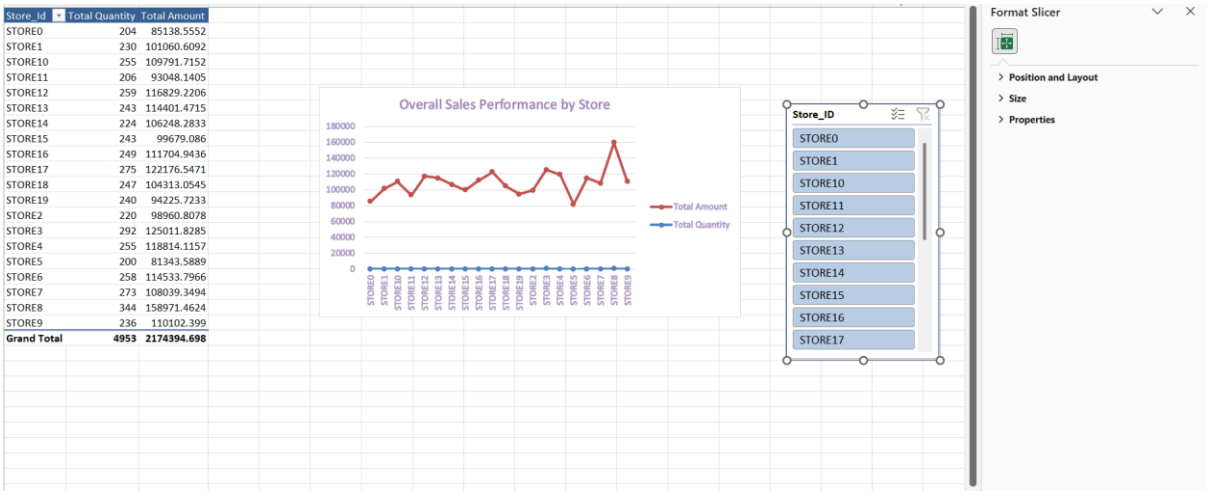
Pivot Tables



Year	Total Quantity	Total Amount
2023	515	223812.7245
2024	2484	1093911.795
2025	1954	856670.1784
Grand Total	**4953**	**2174394.698**



Category	Total_Quantity	Total Amount
Beauty	738	₹3,21,561.33
Clothing	844	₹3,79,216.04
Electronics	1165	₹5,04,311.35
Home	965	₹4,27,295.80
Sports	1241	₹5,42,010.17
Grand Total	**4953**	############



Store_Id	Total Quantity	Total Amount
STORE0	204	85138.5552
STORE1	230	101060.6092
STORE10	255	109791.7152
STORE11	206	93048.1405
STORE12	259	116829.2206
STORE13	243	114401.4715
STORE14	224	106248.2833
STORE15	243	99679.086
STORE16	249	111704.9436
STORE17	275	122176.5471
STORE18	247	104313.0545
STORE19	240	94225.7233
STORE2	220	98960.8078
STORE3	292	125011.8285
STORE4	255	118814.1157
STORE5	200	81343.5889
STORE6	258	114533.7966
STORE7	273	108039.3494
STORE8	344	158971.4624
STORE9	236	110102.399
Grand Total	**4953**	**2174394.698**

Visualization and Insights Arrived with Screenshot of Visualization Report/Dashboard Screenshot

1. Descriptive Analysis (What happened?)

- What are the total sales and total items sold per category?
- Which region generated the highest revenue?
- Which year had the highest sales revenue?

2. Diagnostic Analysis (Why did it happen?)

- Why did revenue experience a downturn relative to the 2024 performance?
- Why does the West region underperform in sales compared to other geographic regions?

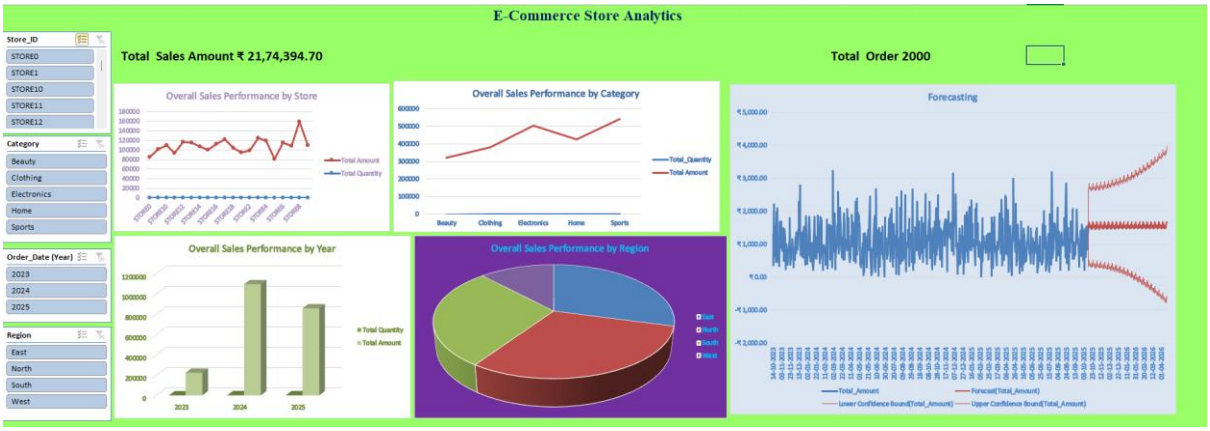
3. Predictive Analysis (What is likely to happen?)

- How many sales do we expect to make next quarter based on past seasons?

4. Prescriptive Analysis (What should we do?)

- How can we increase sales and marketing for our lowest-performing category, Beauty?

Dashboard



Final 4 Type of Insights as Storytelling

Our store earned a total of ₹21,74,394.70 from 4,953 items sold, with Sports and Electronics being our biggest sellers, mostly in the South and East regions. While 2024 was our best year, sales later dipped due to fewer promotions, and the West region fell behind because of fewer stores and lower visibility. Looking ahead, past seasonal patterns show our top items will likely grow by 15% to 20% next quarter. To keep growing, we should stock up on popular

products, improve our marketing in the West, and run special bundle deals to help boost slower categories like Beauty.

Conclusion

Our store earned ₹21,74,394.70 in total sales, driven mainly by Sports and Electronics. By restocking top items, improving West region marketing, and promoting slower categories like Beauty, we are well-positioned for steady growth ahead.